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Hands-on Microsoft Foundry



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Meet Your Instructor – Razi Rais

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- **20+ years** in cybersecurity & systems architecture
- **10+ years** at Microsoft in engineering, architecture, and product leadership
- **Global career** in France, Dubai, Singapore with 15+ years in the United States
- **Applied AI + Cybersecurity** focus applying AI to modern security architectures
- **Author &Trainer** 5 books, Microsoft Certified Trainer (15+ years), GIAC Advisory Board member, RSAC speaker

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Learning Objectives

- Understand what Microsoft Foundry is and how its model catalog, agent runtime, and tools fit together.
- Learn how to build and run agents using Foundry, including system prompts, conversations, and tool integration.
- Explore how Foundry extends agent capabilities through tools such as custom functions, and MCP.
- Gain awareness of Foundry’s observability features for monitoring, tracing, and optimizing agent behavior.
- Show-and-tell approach with demos and interactive Q&A.

Prerequisites

- **No prior Microsoft Foundry experience** is required, though familiarity with Azure Cloud and basic skills in languages like Python or C# are expected.
- We'll cover a lot in this session, but note that it is not an in-depth exploration of every Foundry feature or advanced agent development.
- AI is moving fast, so no one is expected to know everything. Just bring your curiosity!

Resources

- Slides and course recording will be shared after the session. Additional resources and references are available on **GitHub:** razi-rais/microsoft-foundry-resources

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SESSION / REGISTER	DATE	TIME / CENTRAL	WHAT YOU'LL TAKE AWAY
Anatomy of a Cyberattack	**Nov 4 2026**	9 AM to 11 AM CST	Decode bots, identity attacks, and LLM threats. Explore deepfakes and AI supply chain risks.
Agentic AI Security with Microsoft MDASH and Project Perception	**Nov 6 2026**	10 AM to noon CST	Find and validate flaws with MDASH. Explore governed SOC agents with Perception.

All times are US Central. October uses CDT. November uses CST.

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Razi Rais

Your trainer. Author. Security practitioner.

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Poll

Question

How familiar are you with Microsoft Foundry (formerly Azure AI Foundry)? Choose the option that best applies to you.

Choices:

- No experience - I've only heard of it.
- Beginner - I've explored or tested it briefly.
- Intermediate - I've used it in real production projects.
- Expert - I work with it extensively and have command of it.

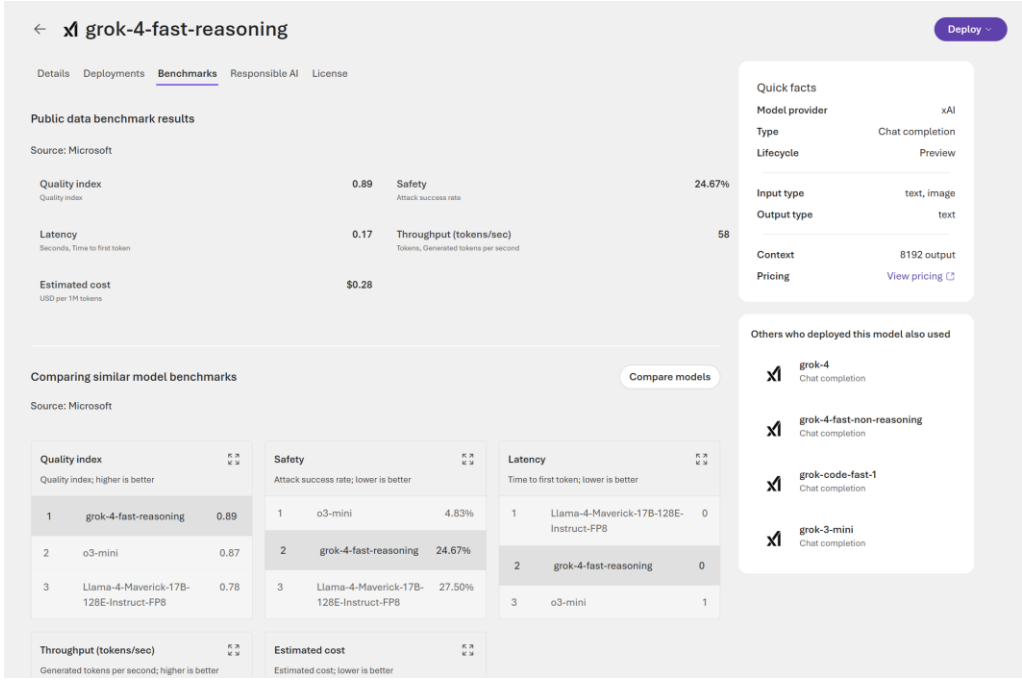
Microsoft Foundry: AI Model Catalog & Offerings

- **Unified AI Model Catalog:** Microsoft provides access to 1900+ models including foundation, reasoning, small language, multimodal, industry-specific, and domain-specific models.
- **Models Sold Directly by Azure**: Microsoft-hosted, enterprise SLAs, deep Azure integration, Responsible AI review.
- **Partner & Community Models:** From Anthropic, Hugging Face, research labs, offering specialized and cutting-edge capabilities.
- Organizational data is never used for training which is critical for building Enterprise AI trust

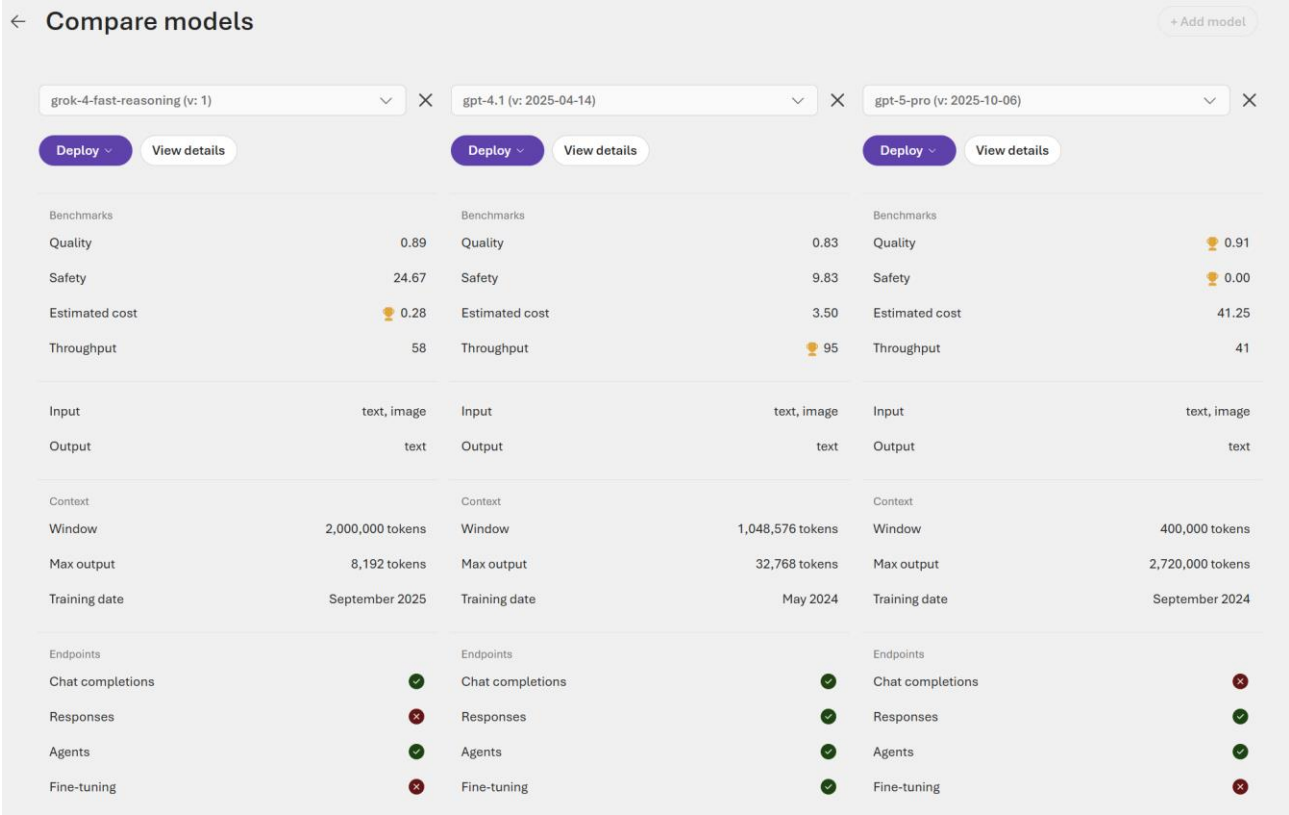
Select models using benchmarks

Compare models by:

- Quality
  - *Quality index* (higher is better)
- Safety
  - *Attack success rate* (lower is better)
- Throughput
  - *Output tokens/sec* (higher is better)
- Cost
  - *USD per 1M tokens* (lower is better)



Compare models



DEMO: Walkthrough of Microsoft Foundry Service



Poll

Question

Which statement best describes Microsoft’s Foundry Model Catalog?

Choices:

- A set of dedicated Microsoft-build models
- A library focused mostly on Microsoft-built models with limited partner options.
- A broad catalog of Microsoft, partner, and community models with strict data privacy.
- A collection that requires users to train models before using them.

DEMO: Exploring LLM by using Playground



Choose your Microsoft Foundry setup

- **Basic Setup**: This setup is quickest and manages agent states using the platform's built-in storage.

- **Standard Setup**: Includes everything in the basic setup and fine-grained control over your data by allowing you to use your own Azure resources. All data including files, threads, and vector stores are stored in your own created Azure resources, giving you full ownership and control.

- **Standard Setup with Bring Your Own (BYO) Virtual Network**: Includes everything in the Standard Setup, with the added ability to operate entirely within your own virtual network. This setup supports Bring Your Own Virtual Network (BYO virtual network), allowing for strict control over data movement and helping prevent data exfiltration by keeping traffic confined to your network environment.

DEMO: Creating a Microsoft Foundry Instance

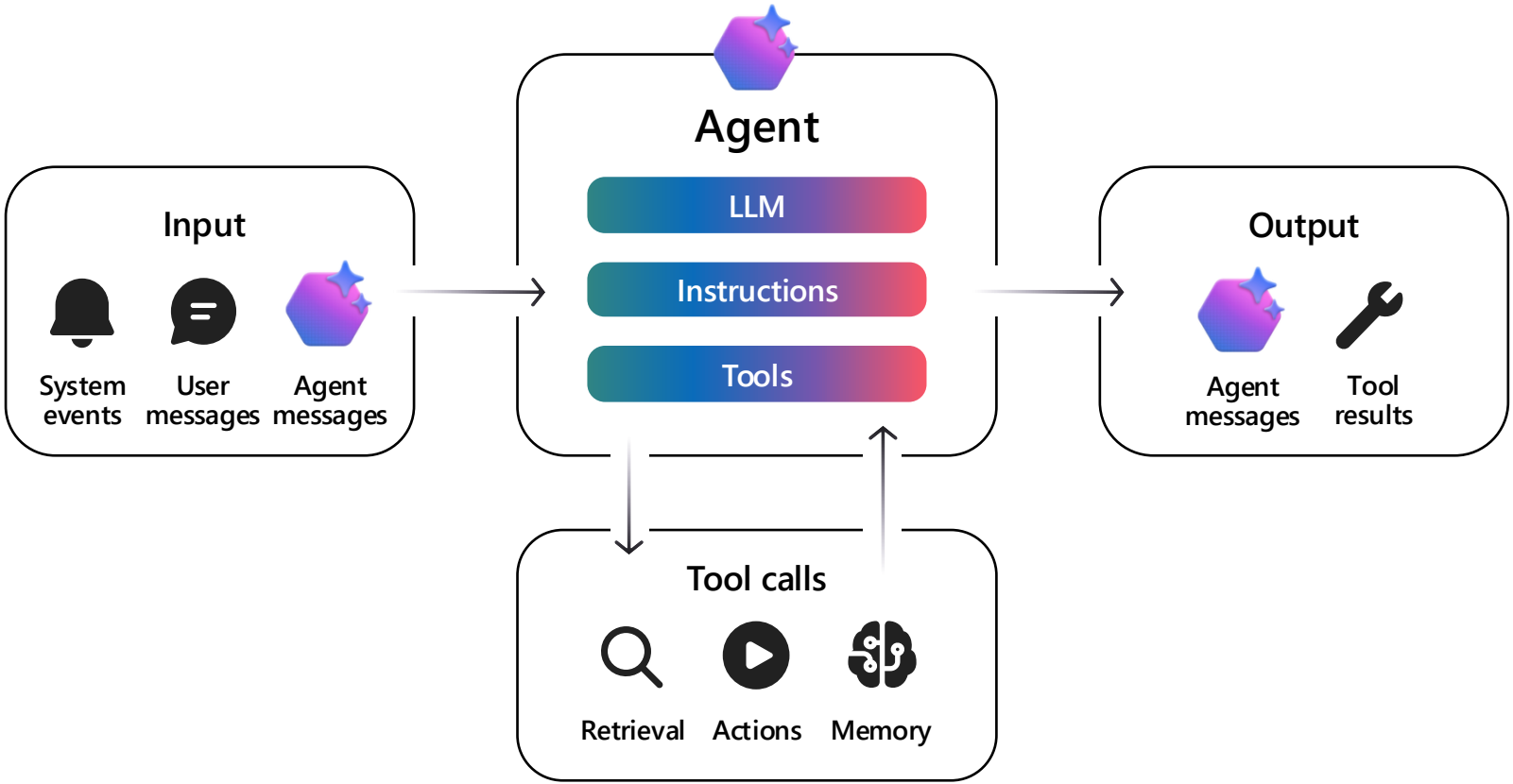


Pulse Check

Do you understand why Microsoft brings models from many sources into Foundry and why it’s important?



What is Microsoft Foundry Agent?



Microsoft Foundry Agent runtime components

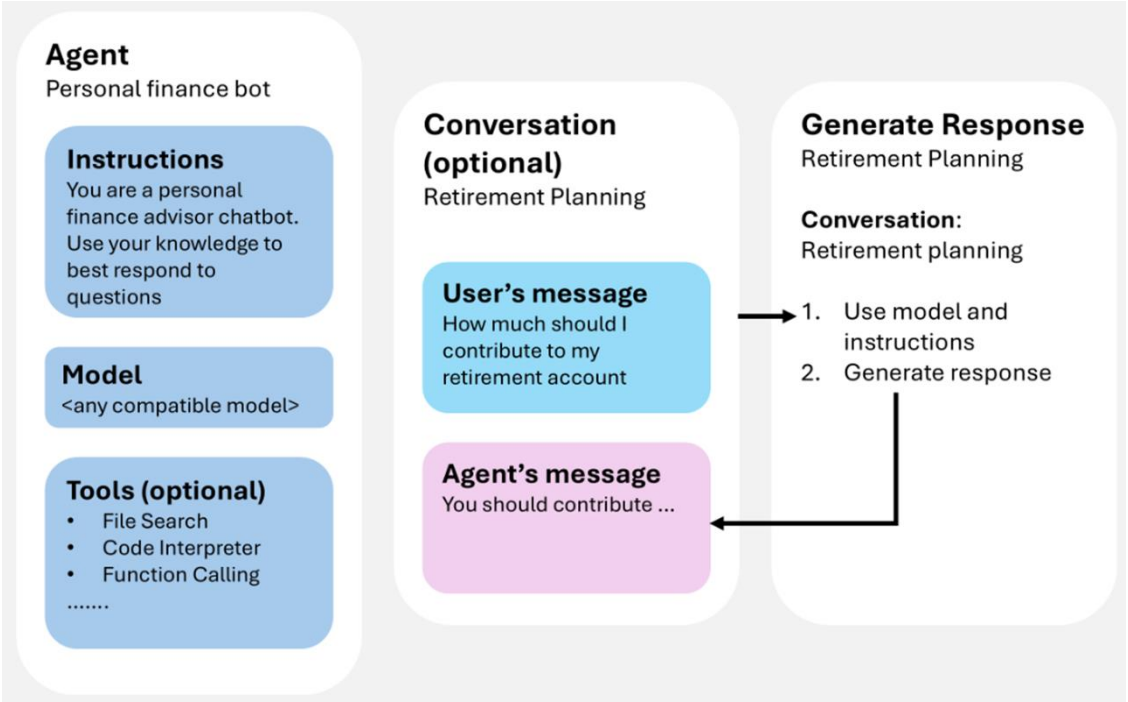
1.Create an agent: Define an agent to start sending messages and receiving responses.

2.Create a conversation (optional): Use a conversation to maintain history across turns. If you don't create one, state is stored automatically with each response.

3.Generate a response: The agent processes input items in the conversation and any instructions provided in the request. Items may be appended to the conversation.

4.Check response status: Monitor the response until it completes (especially in streaming or background mode).

5.Retrieve the response: Display the generated response to the user.

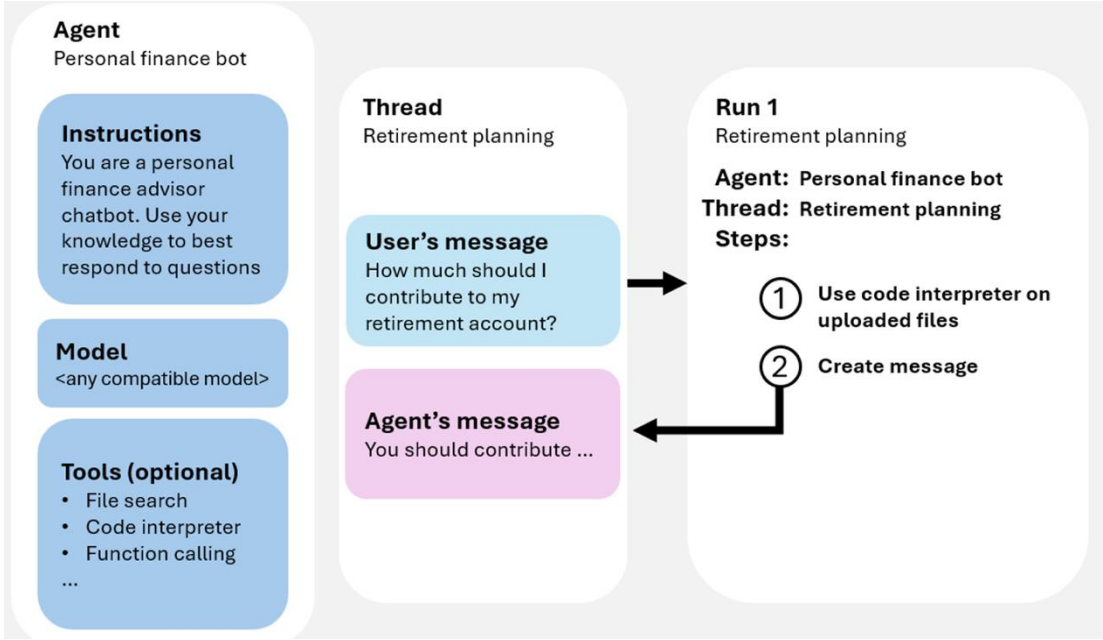


Microsoft Foundry Agent runtime components

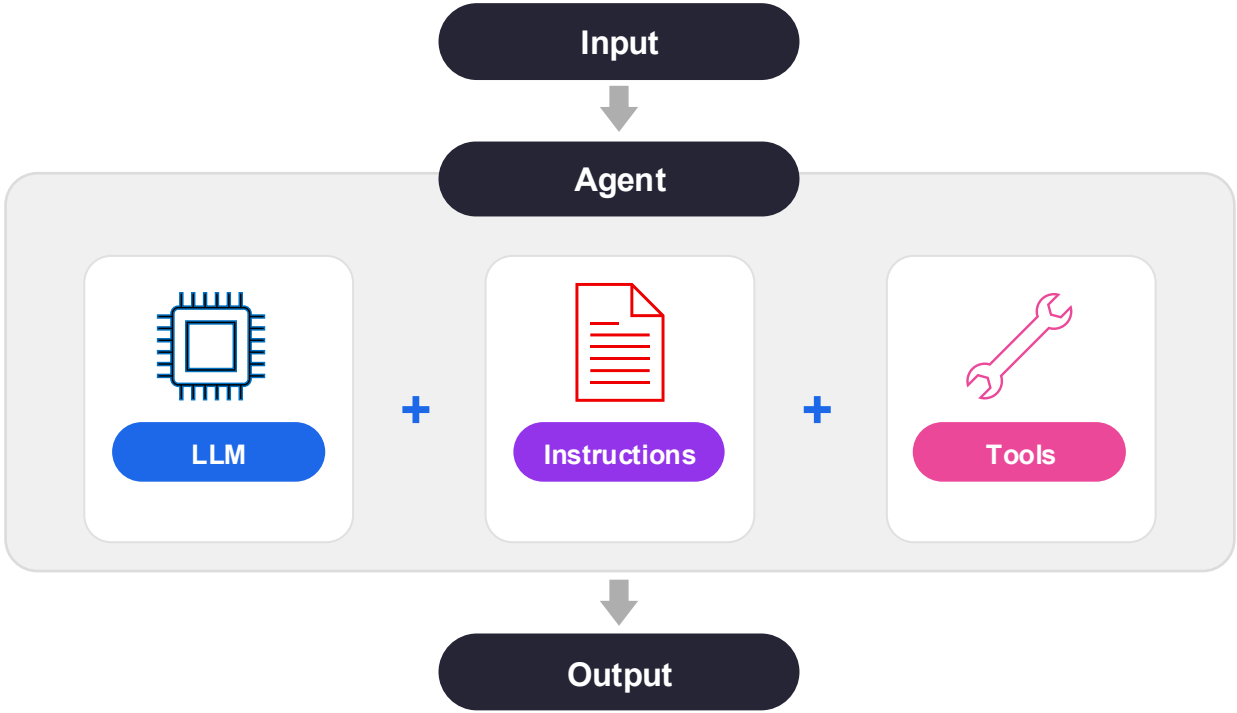
- **Threads**: Threads are conversation sessions between an agent and a user.

- **Messages**: Messages are the individual pieces of communication within a thread. They can be created by either the agent or the user and can include text, or other files. Messages are stored as a list within the thread, allowing for a structured and organized conversation flow.

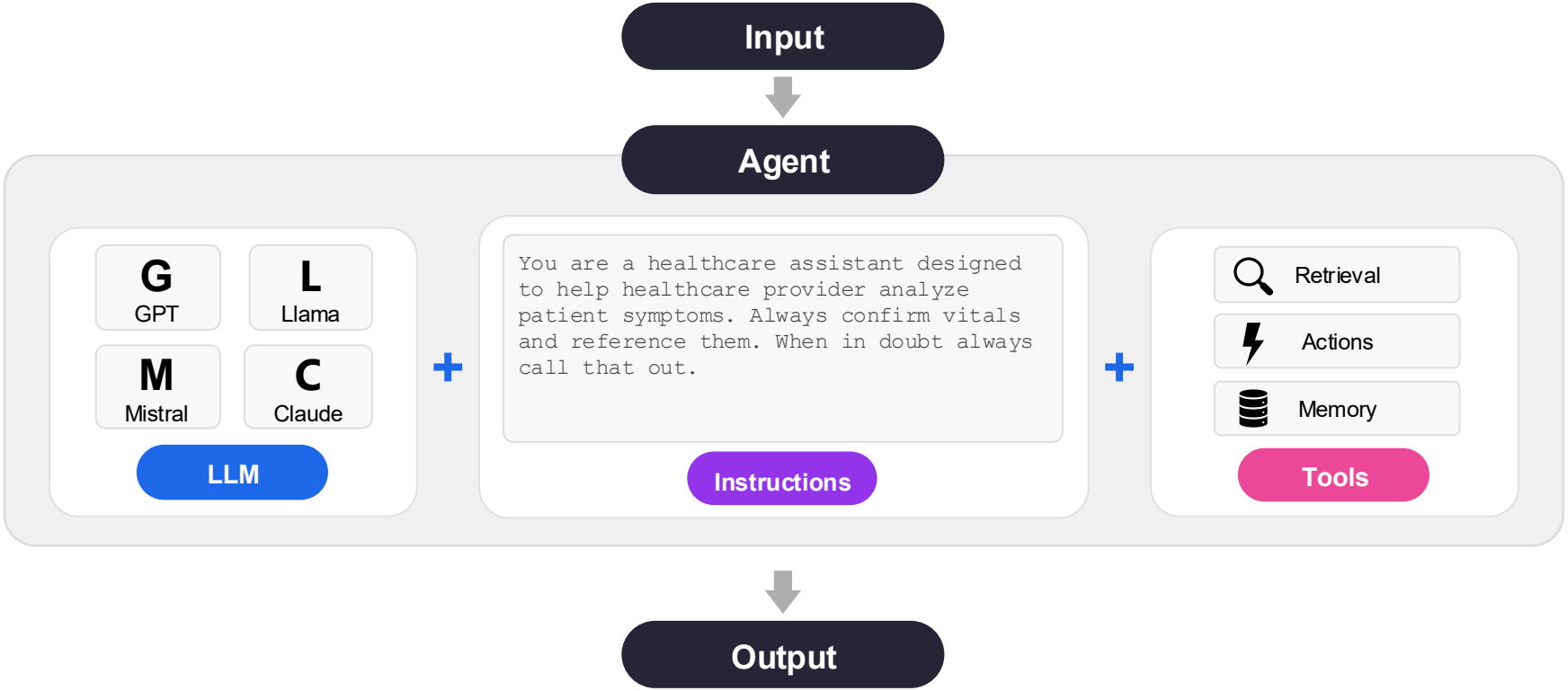
- **Runs**: A run involves invoking the agent on the thread, where it processes the messages in the thread and may append new messages (responses from the agent). The agent uses its configuration and the thread's messages to perform tasks by calling models and tools. As part of a run, the agent appends messages to the thread.



AI Agent



AI Agent


Input
Agent
G
GPT
L
Llama
M
Mistral
C
Claude
LLM
+
You are a healthcare assistant designed to help healthcare provider analyze patient symptoms. Always confirm vitals and reference them. When in doubt always call that out.
Instructions
+
Retrieval
Actions
Memory
Tools
Output

DEMO: Working with agent using Microsoft Foundry portal



DEMO: Working with agent using VS Code



Understanding Microsoft Foundry Tools

- Enable AI agents to take goal-directed actions by calling and interacting with external systems.
- Are invoked based on model output during agent run (non-deterministic)
- Provide the horsepower of enterprise AI by bridging the gap between data and the model.
- **Knowledge tools:** Expand an AI agent's understanding by supplying richer context from diverse data sources private, public, licensed, structured, and unstructured so the agent can stay well-informed.
- **Action tools**: Enable the agent to execute workflows and take real-world actions through options ranging from no-code integrations to custom functions, APIs, MCP endpoints, deep research pipelines, and browser automation.

Build-in Microsoft Foundry Tools

Tool	Description
Azure AI Search	Use an existing Azure AI Search index to ground agents with data in the index, and chat with your data.
Azure Functions	Leverage your Azure Functions to create intelligent, event-driven applications.
Browser Automation	Perform real-world browser tasks through natural language prompts.
Code Interpreter	Enable agents to write and run Python code in a sandboxed execution environment.
Deep Research (preview)	Use OpenAI's advanced agentic research capability for analysis and reasoning.
File Search	Augment agents with knowledge from outside its model, such as proprietary product information or documents provided by your users.
Function calling	Describe the structure of functions you create to an agent and have them be called when appropriate during the agent's interactions with users.
Grounding with Bing Search	Enable your agent to use Grounding with Bing Search to access and return information from the internet.
Grounding with Bing Custom Search (preview)	Enhance your Agent response with selected web domains
Model Context Protocol (preview)	Give the agent access to tools hosted on an existing MCP endpoint
Microsoft Fabric (preview)	Integrate your agent with the Microsoft Fabric data agent to unlock powerful data analysis capabilities.
OpenAPI 3.0 Specified tool	Connect your Azure AI Agent to external APIs using functions with an OpenAPI 3.0 specification.

Non-Microsoft Foundry Tools

Tool	Description
Auquan	AI-powered workflow automation for institutional finance
Celonis	Celonis delivers Process Intelligence to accelerate enterprise AI at scale
InsureMO Insurance Quotation	Action APIs for insurance quotations for Car, Home, and Travel
LEGALFLY	Legal insights grounded in trusted sources from your jurisdiction.
LexisNexis	Seamless access to LexisNexis content.
MiHCM	seamless integration with MiHCM's HR functionalities
Morningstar	Access up-to-date investment research and data such as analyst research, expert commentary, and essential Morningstar data.
Trademo	Provide latest duties and past shipment data for trade between multiple countries
Tripadvisor	Get travel data, guidance and reviews
Tool	Description
Auquan	AI-powered workflow automation for institutional finance
Celonis	Celonis delivers Process Intelligence to accelerate enterprise AI at scale

Pulse Check

Do you feel clear on the role of tools within Microsoft Foundry



DEMO: Adding Tools to the Agent



Poll

Question

Which statement best describes what tools do for AI agents?

Choices:

- Tools let agents access read-only data but cannot initiate any actions.
- Tools supply structured datasets but do not interface with external systems.
- Tools let agents access data and take actions in external systems.
- Tools only allow local, pre-approved operations with no workflow execution.

DEMO: Working with Tools
Custom Functions



Comparing approaches

Foundry portal

Visual, web-based, no local setup required

- ✓ **Quick prototyping**
  Test agent concepts without setting up a local development environment
- ✓ **Visual configuration**
  Configure agents through intuitive forms and dropdowns rather than code
- ✓ **Centralized management**
  View and manage all agents across projects in one place with dashboards
- ✓ **Team collaboration**
  Share configurations with stakeholders who prefer visual interfaces

Visual Studio Code

Developer-centric, code-integrated, Git-friendly

- ✓ **Developer-centric workflows**
  Build agents alongside application code in a single familiar environment
- ✓ **Version control integration**
  Track agent configurations in Git alongside your codebase
- ✓ **Code-first development**
  Edit YAML configurations directly and integrate agents into application logic
- ✓ **Local development**
  Design and test agents offline before deploying to Azure

Many teams use both: the Foundry portal for initial exploration and reviews, and VS Code for detailed development and production deployments.

Model Context Protocol (MCP)

- A **standardized protocol** that lets AI agent access tools, data sources, and systems in a consistent, secure way.

- Acts as a **universal bridge** between LLMs based agents and external applications, APIs, files, and databases.

- Provides **structured, context-rich interactions** such agents can understand, retrieve, and act on information more reliably.

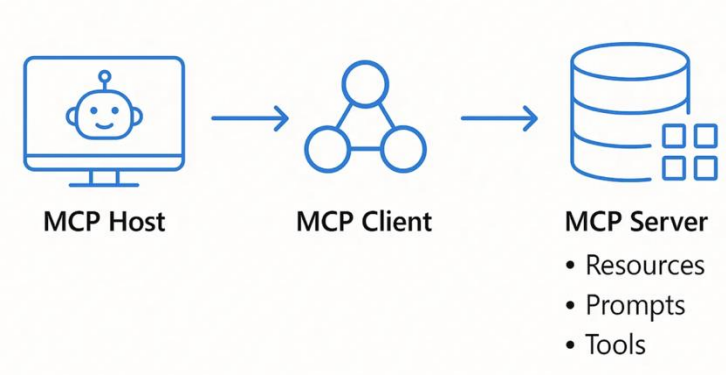
Model Context Protocol (MCP) – Terminology

• MCP operates on a client-server architecture and the core components are:

• **Hosts** are LLM applications (for example a code editor like VS Code) that start the connections to an MCP Server.

• **Clients** are components within the host application that maintain one-to-one connections with servers.

• **Servers** are lightweight programs that expose specific capabilities.



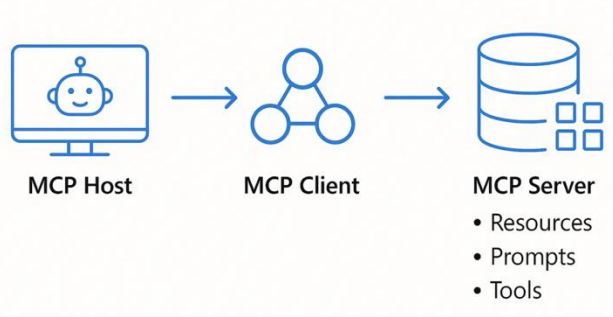
Model Context Protocol (MCP) – Terminology (cont.)

Tools: These are discrete actions or functions an AI agent can call to perform an action. For example, a weather service might expose a "get weather" tool, or an e-commerce server might expose a "purchase product" tool. MCP servers advertise each tool's name, description, and input/output schema in their capabilities listing.

Resources: These are read-only data items or documents that an MCP server can provide, and clients can retrieve them on demand. Examples include file contents, database records, or log files. Resources can be text (like code or JSON) or binary (like images or PDFs).

Prompts: These are predefined templates that provide suggested prompts, allowing for more complex workflows.

NOTE: Enterprises increasingly expose existing service endpoints (e.g., APIs) as tools rather than as prompts or static resources, because this makes them far easier and faster to integrate



Model Context Protocol (MCP) – Key Benefits

• **Dynamic Tool Discovery**: Agents can dynamically receive a list of available tools from a server along with descriptions of what they do. This contrasts with traditional APIs, which often require static coding for integrations, meaning any API change necessitates code updates. MCP offers an "integrate once" approach, leading to greater adaptability.

• **Interoperability Across LLMs**: MCP works across different LLMs, providing flexibility to switch core models to evaluate for better performance.

Integrating MCP with Foundry – Key Components

MCP Server: The MCP server acts as a registry for tools your agent can use.

MCP Client: A standard MCP client acts as a bridge between your MCP server and the Agent Service

How it works:

1. The **MCP server** hosts tool definitions decorated with @mcp.tool.
2. The **MCP client** initializes an MCP client connection to the server.
3. The **MCP client** fetches the available tool definitions with session.list_tools().
4. Each tool is wrapped in an async function that invokes session.call_tool
5. The tool functions are bundled into FunctionTool that makes them usable by the agent.
6. The FunctionTool is registered to the agent's toolset.

Dynamic Tool Discovery using MCP

Dynamic tool discovery is a mechanism allows Foundry AI agent to discover available external tools by querying a centralized Model Context Protocol (MCP) server. This server acts as a live catalog, exposing tools that the agent can understand and call.

• Tools can be added, updated, or removed centrally without modifying the agent code.

• Agents can always use the latest version of a tool, improving accuracy and reliability.

• The complexity of managing tools shifts away from the agent and into a dedicated service.

Pulse Check

Does the role of MCP in AI agent development make sense so far?



Poll

Question

Which of the following best describes what the MCP (Model Context Protocol) provides?

Choices:

- A protocol that only handles simple message passing between LLMs.
- A system that standardizes how AI agents store their internal memory externally.
- A standardized protocol that allows AI agents to external systems in a consistent, secure way.
- A mechanism that limits AI agents to interacting only with local files on a single machine.

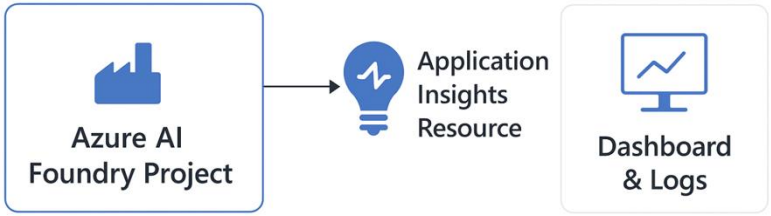
DEMO: Connect AI agent to tools using Model Context Protocol (MCP)



Observability in Microsoft Foundry

- Built-in observability with Application Insights provides continuous insight into model performance, token usage, and prompt behavior, helping ensure AI applications stay reliable in production.

- Fine-grained, code-level agent tracking through OpenTelemetry provides detailed metrics on quality, performance, safety, and cost for faster diagnosis and optimization.



DEMO: Enabling Monitoring In Microsoft Foundry



Tracing in Agents

- Agent runs can be complex, with many steps, varying paths, and long intermediate inputs and outputs that make issues hard to track.
- Tool calls deepen that complexity, making it difficult to pinpoint where errors or unexpected behavior originate.
- Trace solve this by revealing every step’s inputs and outputs in order, providing the visibility needed to diagnose and debug agent behavior.

DEMO: Adding Tracing Capability using OpenTelemetry



Q&A



Takeaways

- You gained an understanding of how Microsoft Foundry unifies models, tools, agent id, MCP, and observability into an enterprise-ready platform for building AI agents
- AI is evolving rapidly, so stay current by using following resources!

- **Access free 1-hour** on-demand or attend live webinars posted on LinkedIn: https://www.linkedin.com/in/razirais
- **O'Rielly Trainings**: Live Training Courses (AI + Cybersecurity)
- Don't forget to check **GitHub for resources**: razi-rais/microsoft-foundry-resources
- Feel free to connect for expert advice or collaboration.
  - LinkedIn: https://www.linkedin.com/in/razirais
  - Meeting: https://calendly.com/razi-rais/30min

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